

Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

Data Science Center, UCLA Library

UCLA OSPO Lead, UC OSPO Network

The invisible layer

Research data workflows run on open source:

- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

92% of researchers use software; **69%** say their research would not be practical without it.

The people keeping it running are increasingly called

Research Software Engineers (RSEs), a growing professional community with no standard institutional home.

If every organization had to replace open source with proprietary equivalents, the cost would exceed **\$8.8 trillion**.

Hettrick et al. (SSJ, 2014/2022) ·
Hoffmann, Nagle & Zhou (2024)
doi:10.2139/ssrn.4693148



Jupyter



Python



R



Git



Spark



Pandas



NumPy



Docker



Ray



ggplot2



dplyr



Snakemake



Nextflow

The UC research-to-infrastructure pipeline

Data professionals built open data infrastructure: FAIR, DMPs, curation, provenance. **FAIR4RS** is the same inflection point for software — and UC has been showing the way for 50 years.

BSD Unix · Ceph · Apache Spark · Jupyter · RISC-V · Ray

The pipeline from university research to global infrastructure is not rare or accidental.

Barker et al. (2022)
doi:10.1038/s41597-022-01710-x

UC Berkeley |



Jupyter



Apache Spark



RISC-V



Ray



BSD Unix

UC Santa Cruz |



Ceph

UCSC Genome Browser

UC San Diego |

Cytoscape

EEGLAB



UCSD Pascal

UCLA |



Processing



Named Data Networking

tz database

UC San Francisco |

ChimeraX



OpenMM

UC Davis |



sourmash

ERPLAB

What is an academic OSPO?

An Open Source Program Office

supports, governs, and promotes open source within an institution.

- Originated in industry:
~77% of large tech companies have one
- First academic OSPO:
Johns Hopkins, 2019
- **34 academic OSPOs** in CURIOSS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



● UC OSPO Network · ● CURIOSS member

World map showing 34 CURIOSS member institutions. Six U

The UC OSPO Network

A **multi-campus collaboration** treating open source as shared infrastructure.

- 6 campuses: Santa Cruz, Berkeley, Davis, Los Angeles, Santa Barbara, San Diego
 - Launched April 2024, funded by the Alfred P. Sloan Foundation
 - **\$1.85M** in external funding to date
 - Lead: UC Santa Cruz, the first OSPO in a large state university system
 - Serves 280,000+ students and 25,000+ faculty
- Acts as a **neutral convener**: no single campus, company, or grant

cycle can pull the work
away.

Ruff (2026) UC Open
Summit ·

youtu.be/eBriL3CDNeo



Map of California showing six UC OSPO Network campuses:

Discovery · Sustainability · Education



Discovery

Mapping who
does what
across the UC
system

- 236,000+
repos
scanned;
~82,000
institutionally
affiliated
- 294-
respondent
system-wide
survey
- UC Open
Repository
Browser (UC
ORB)



Sustainability

Keeping
projects and
communities
healthy

- **58%** of UC contributors are also maintainers
- Cross-campus licensing working group (UCOP + tech transfer)
- Project health assessment (OSSPREY)



Education

Coordinated
training across

campuses

- Gap analysis
→ curriculum
inventory →
learning
pathways
- Published at
ucospo.net/education
- Coordinated
with
Carpentries
infrastructure

Gomez et al. (2025) arxiv:2506.18359 · Scarlett et al. (2025) doi:10.31235/osf.io/p8bx6_v1

Lessons from two years

What the network model enables that single campuses cannot:

- .. **Shared staffing:** community manager, licensing specialist, technical roles
- 2. **Policy leverage:** the network has standing with UCOP; individual campuses do not
- 3. **Data at scale:** the GitHub and survey analyses require system-wide scope to mean anything
- 4. **Equity:** smaller campuses get services they could not build alone
- 5. **A replicable model:** \$1.85M produced something other state systems can follow

Each campus contributes unique expertise; the network routes it to where it's needed.

Where this meets your work

For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

For your institution:

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?

You're already doing this work

If you do technical data services, researchers are already bringing you these questions:

- “What license should I use for this code?”
- “Is this library still maintained? Should I depend on it?”
- “How do I make this reproducible and citable?”
- “My funder wants a software management plan.”
- “How do I accept contributions from collaborators?”

That is OSPO work. The frame makes it legible to your institution.

The network has already built the resources (Laura Langdon, UC OSPO):

[README](#) · [LICENSE](#) · [CONTRIBUTING](#) ·

CODE OF CONDUCT ·

CITATION.cff + Zenodo DOI

· **Growing a Community** ·
Carpentries Incubator
lesson**Raise the floor, not
just solve the ticket**

Fixing their immediate
problem gets them
through the week.

Moving them from “it
runs on my machine” to
maintainable, licensed,
citable software gets
their whole lab there,
and their students after
them.

You don’t need a formal
OSPO to do this. You
need the frame.

**OSPO heuristics
already in your toolkit:**
health signals · bus
factor · governance files
· FAIR4RS · software
DMPs · contributor

conventions · community
governance

Learn more / connect

UC OSPO Network: ucospo.net

- Education site: ucospo.net/education
- Mailing list, Slack, events calendar

Global network: curiOSS.org, 34 academic OSPOs worldwide and growing

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Slides: tinyurl.com/iassist2026



Handout:

tinyurl.com/iassist2026-handout



QR code linking to
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slides for this presentation

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Speaker notes